IT Project Documentation

Version 1.0

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Project: Creating IAM custom policy and User

Step1: Generating the JSON policy through AWS Policy generator

1. Let's open the AWS Policy Generator by going to *https://awspolicygenerator.com*

2. Then, lets' select Policy Type

From **Select Type of Policy**, choose **AWS Identity and Access Management (IAM) Policy**.

3. Add Statements

Effect: Choose Allow or Deny.

AWS Service: Pick the service (e.g., S3, EC2, Cloudwatch, Lambda).

Actions: Select the specific actions (e.g., here we'll select all).

ARN (Amazon Resource Name): Specify the resource(s) the policy applies to (we'll apply to all).

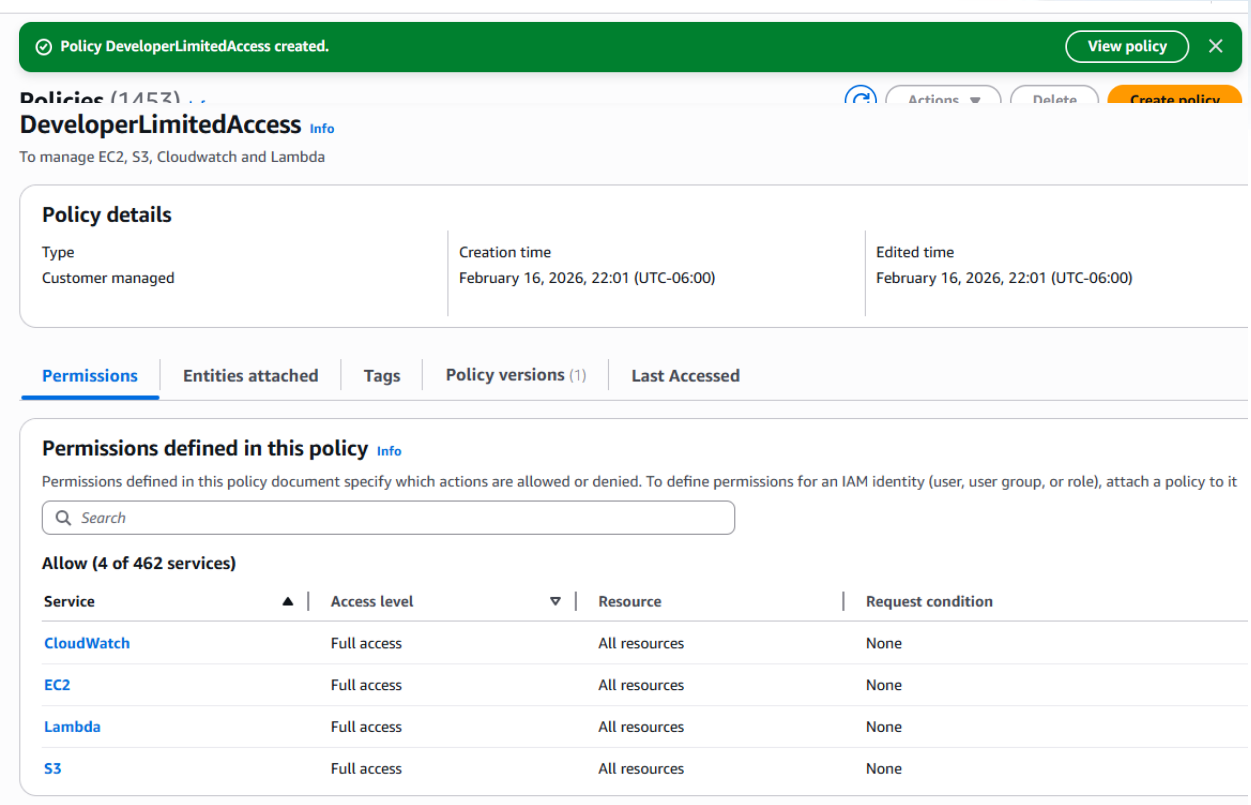
4. Add More Statements (one per service, so 4 totally)

Click Add Statement to include multiple permissions in one policy.

5. Generate Policy

Once all statements are added, click **Generate Policy**.

The tool will produce the following JSON-formatted IAM policy.



Step 2: Create the IAM user in the Console

1. Sign in to AWS Management Console
Let's Go to the AWS Management Console and sign in with our admin account credentials.

2. Navigate to IAM Service
In the search bar, type "IAM" and select the IAM service from the dropdown.

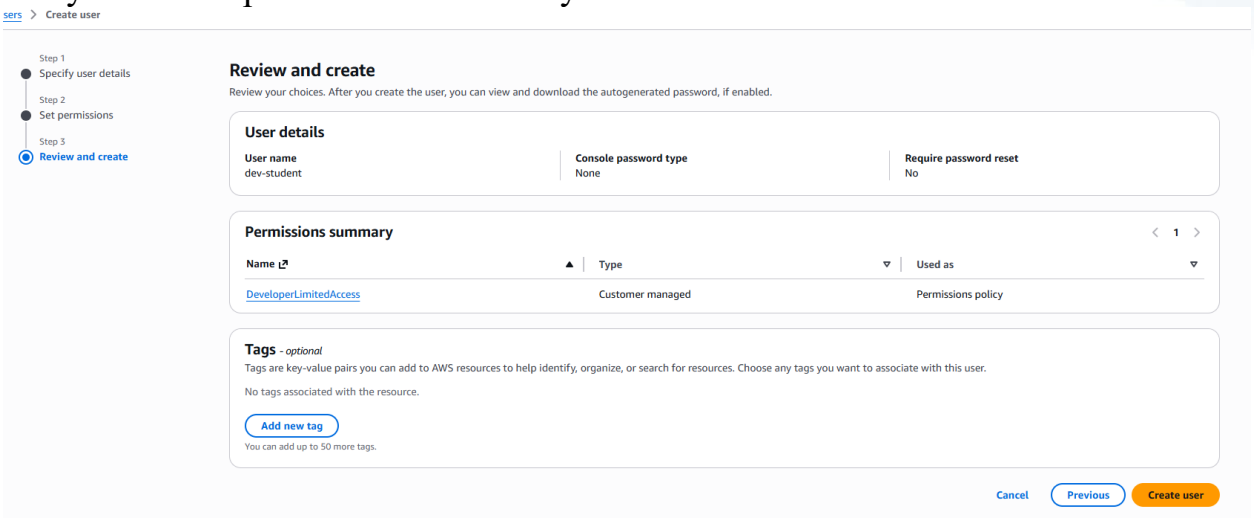
3. Create a New User
Click on "Users" in the left-hand menu, then click on the "Add user" button.

4. Specify User Details
Enter a username for the new user. Select the type of access this user will have.
Programmatic access: Provides access via the AWS CLI.

5. Set Permissions
Attach policies directly: Attach the previously created policy directly to the user.

6. Review and Create

Review your settings and click "Create user". The new IAM user will be created, and you will be provided with security credentials.



Step 3: Create the access key

1. In the IAM dashboard, click on Users in the left-hand menu.
2. Select the IAM user for whom you want to create the access key.
3. Navigate to the **Security credentials** tab within the user's details page.
4. Under the **Access keys** section, click on **Create access key**.

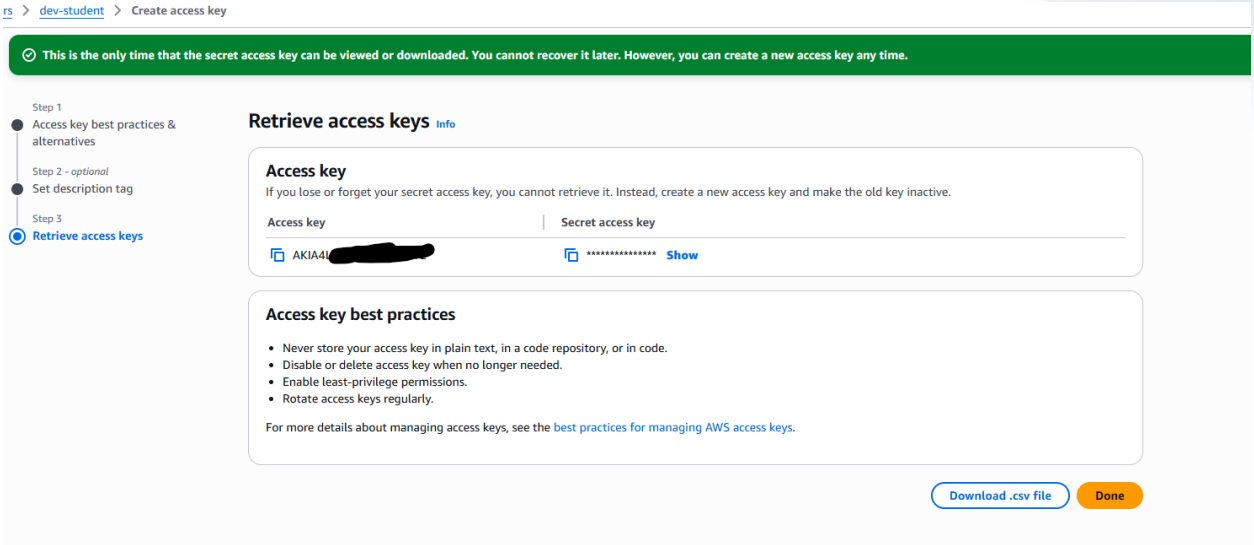
A dialog box will appear displaying the newly generated Access Key ID and Secret Access Key.

5. Click on **Download .csv file** to save the keys securely to your local system.

Ensure you store this file in a secure location, as the Secret Access Key cannot be retrieved again after this step.

Important Notes:

- Do not share your access keys or upload them to public repositories.
- If the keys are compromised, delete them immediately and create new ones.
- Use the keys for programmatic access to AWS resources via AWS SDKs, CLI, or other tools.



Step 4: Configure your AWS CLI

1. Open Windows command prompt.
2. Enter 'aws configure' command to configure the API access.
3. Provide the access key ID, secret access key , default region and default output format.

```
C:\Users\Admin>aws configure
AWS Access Key ID [****************LQHM]: AKIA[redacted]
AWS Secret Access Key [****************RNmy]: XFeIDg[redacted]
Default region name [us-east-1]:
Default output format [json]:
```

Step 5: Test AWS CLI commands to confirm the user has the right access.

1. Testing EC2 permissions

```
C:\Users\Admin>aws ec2 describe-instances
{
    "Reservations": [
        {
            "ReservationId": "r-068d[redacted]4cbe",
            "OwnerId": "84[redacted]15",
            "Groups": [],
            "Instances": [
                {
                    "Architecture": "x86_64",
                    "BlockDeviceMappings": [
                        {
                            "DeviceName": "/dev/xvda",
                            "Ebs": {
                                "AttachTime": "2023-07-25T00:30:57+00:00",
                                "DeleteOnTermination": true,
                                "Status": "attached",
                                "VolumeId": "vol-09[redacted]dea0f",
                                "EbsCardIndex": 0
                            }
                        }
                    ],
                    "ClientToken": "9[redacted]-0[redacted]6-ad51[redacted]",
                    "EbsOptimized": false,
                    "EnaSupport": true,
                    "Hypervisor": "xen",
                    "NetworkInterfaces": [
```

2. Testing S3 permissions

```
C:\Users\Admin>aws s3 ls
2024-04-11 07:48:45 aws-cloudtrail-logs-8[redacted]41d
2023-02-18 12:48:58 elasticbeanstalk-[redacted]east-1-8[redacted]
2024-08-02 19:00:00 si[redacted]iz
2023-07-10 19:55:43 www.conceptix-art.com
```

3. Testing Cloudwatch permissions

```
C:\Users\Admin>Aws cloudwatch list-metrics
{
    "Metrics": [
        {
            "Namespace": "AWS/Usage",
            "MetricName": "CallCount",
            "Dimensions": [
                {
                    "Name": "Type",
                    "Value": "API"
                },
                {
                    "Name": "Resource",
                    "Value": "ListClusters"
                },
                {
                    "Name": "Service",
                    "Value": "ECS"
                },
                {
                    "Name": "Class",
                    "Value": "None"
                }
            ]
        },
        {
            "Namespace": "AWS/Usage",
            "MetricName": "CallCount",
            "Dimensions": [
                {
                    "Name": "Type",
                    "Value": "API"
                }
```

4. Testing Lambda permissions

```
C:\Users\Admin>aws lambda list-functions
{
    "Functions": [
        {
            "FunctionName": "myweek10sqslambda",
            "FunctionArn": "arn:aws:lambda:us-east-1.[redacted]nction:myweek10sqslambda",
            "Runtime": "python3.8",
            "Role": "arn:aws:iam:[redacted]_SQS_Lamba",
            "Handler": "lambda_function.lambda_handler",
            "CodeSize": 299,
            "Description": "",
            "Timeout": 3,
            "MemorySize": 128,
            "LastModified": "2023-06-22T03:29:48.360+0000",
            "CodeSha256": "fI06ZlRH/k[redacted]=",
            "Version": "$LATEST",
            "TracingConfig": {
                "Mode": "PassThrough"
            },
            "RevisionId": "ac9e4471-4baa-43e[redacted]",
            "PackageType": "Zip",
            "Architectures": [
                "x86_64"
            ],
            "EphemeralStorage": {
                "Size": 512
```